

JECRC UNIVERSITY
School of Engineering
B. Tech; I Semester (Common to all)
Subject: Mathematics I (DMA024A)
Tutorial Sheet - 15

Time: 2 hrs.

Max. Marks: 64

Instructions:

1. Attempt all the questions in sequence.
 2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
 3. Write the relevant question number before writing the answer.
- CO5: To acquaint the student with mathematical tools needed in evaluating multiple integrals and their usage.

Part A

(5×2=10)

- A1. [CO5] Find the angle between the vectors $\vec{a} = 3i - 4j$ and $\vec{b} = 6i - 3j + 2k$. **Ans.:2/25**
- A2. [CO5]) If $\vec{r} = xi + yj + zk$, then find $\nabla \cdot \vec{r}$ is **Ans.:3**
- A3. [CO5] If $\vec{r} = 3i - j + 2k$ and $\vec{s} = i - 2j - k$, find the value of $\vec{r} \times \vec{s}$. **Ans.:5j-5k**
- A4. [CO5] Find the directional derivative of $f(x,y,z) = xy^2 + yz^3$ at the point(2,-1,1)in the direction of the vector $i + 2j + 2k$. **Ans.: -11/3**
- A5. [CO5] Prove that the vector field defined by

$$\vec{V} = (x^2 + xy^2)i + (y^2 + x^2y)j \text{ is irrotationqal.}$$

Section B

(3x7=21 Marks)

- B.1 [CO5] A particle moves along the curve $\vec{r} = (t^3 - 4t)i + (t^2 + 4t)j + (8t^2 - 3t^3)k$, where t denotes time.Find the magnitude of acceleration along the tangent and normal at time t=2. **Ans.: 16,2√73**

B.2 [CO5]] $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$ where $r^2 = x^2 + y^2 + z^2$

- B.3[CO5] Find the value of λ which makes the vectors $i - j + k, 2i + j - k$ and $\lambda i - j + \lambda k$ coplanar.

Section C

(3x11=33 Marks)

- C.1 [CO5] Using Green's Theorem, evaluate $\int y x^2 dx + x^2 dy$, where c is the boundary describe counter clockwise of the triangle with vertices (0,0),(1,0),(1,1). **Ans.5/12**
- C.2 [CO5] Evaluate $\oint \vec{F} \cdot d\vec{r}$ by Stoke's Theorem, where $\vec{F} = (x^2 + y^2)i - 2xyj$ and C is the boundary of the rectangle $x=\pm a, y = 0$ and $y = b$. **Ans. -4ab²**
- C.3. [CO4] State Gauss Divergence theorem $\int \int \vec{F} \cdot \hat{n} ds = \iiint \text{Div } \vec{F} dv$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 16$ and $\vec{F} = 3xi + 4yj + 5zk$. **Ans.3584π/3**